Career Launcher MBA

QA Arithmetic – 12

Clocks; Boats and Streams; Escalators

Question 1

It takes the pendulum of a clock 7 seconds to strike 4 o’clock. How much time will it take to strike 11 o’clock?

(1) 18 seconds

(2) 20 seconds

(3) 19.25 seconds

(4) 23.33 seconds

Question 2

The time between 7 and 8 o'clock, correct to the nearest minute, when the hands of a clock will form an angle of 84 degrees are:

(1) 7 : 23 and 7 : 53

(2) 7 : 20 and 7 : 50

(3) 7 : 22 and 7 : 53

(4) 7 : 23 and 7 : 52

Question 3

Between 6 PM and 8 PM when are the hands of a clock separated by

(1) 30°

(2) 0°

Question 4

Hour hand and minute hand of a clock coincides every 64 minutes. How much time does the clock gain or lose in a day?

(1) $32\frac{8}{11}$ minutes gain

(2) $32\frac{8}{11}$ minutes lose

(3) $31\frac{4}{11}$ minutes gain

(4) $31\frac{4}{11}$ minutes lose

Question 5

In a watch, the minute hand crosses the hour hand for the third time exactly after every 3 hr 18 min and 15 s of watch time. What is the time gained or lost by this watch in one day?

(1) 14 min 10 s lost

(2) 13 min 50 s lost

(3) 13 min 20 s gained

(4) 14 min 40 s gained

Question 6

A left home between 2 and 3 and came back between 3 and 4. If hour and minute hand exchanged position, for how much approximate time did A go out (in minutes)?

(1) 55

(2) 51

(3) 60

(4) 68

Question 7

A boat sails m km upstream at the rate of r km/hr. If the stream flows at the rate of s km/hr, then how long will the boat take for the return trip?

(1) $\frac{m}{r+2s}$ hr

(2) $\frac{r+2s}{m}$ hr

(3) $\frac{m}{r+s}$ hr

(4) None of these

Question 8

A boat goes from point A to B in upstream and returns to A through the same path. If time of journey from A to B is twice that of from B to A, then the ratio of speed of the boat in still water to the speed of the current is:

(1) 3 : 1

(2) 1 : 3

(3) 2 : 1

(4) Cannot be determined

Question 9

At his usual rowing rate, Rahul can travel 12 miles downstream in a certain river in 6 hr less than it takes him to travel the same distance upstream. But if he could double his usual rowing rate for this 24 miles round trip, the downstream 12 miles would then take only 1 hr less than the upstream 12 miles. What is the speed of the current in miles per hour?

(1) $\frac{7}{3}$

(2) $\frac{4}{3}$

(3) $\frac{5}{3}$

(4) $\frac{8}{3}$

Question 10

A swimmer jumps from a bridge over a canal and swims 1 km upstream. After that first kilometer, he passes a floating cork. He continues swimming for another half an hour and then turns around and swims back to the bridge in 30 mins. The swimmer and the cork arrive at the bridge at the same time. The swimmer has been swimming with constant speed. How fast does the water in the canal flow?

(1) 0.5 km/hr

(2) 2 km/hr

(3) 3.5 km/hr

(4) 1 km/hr

Question 11

A and B are stationed at two points Q and R respectively on a flowing river. The direction of flow of river is from Q to R. When A and B swim towards each other, they meet at P, which is at a distance of 20 m from the point R. When A swims towards B and B swims away from A, they meet at S, which is at a distance of 40 m from the point R. B's speed in still water is five times the speed at which the river flows. Find the ratio of speeds of A and B in still water.

(1) 6 : 1

(2) 7 : 1

(3) 8 : 1

(4) 9 : 1

For Questions 12 and 13

Answer the following questions based on the information given below.

Cities A and B are in different time zones. A is located 3000 km east of B. The table below describes the schedule of an airline operating non-stop flights between A and B. All the times indicated are local and on the same day.

Departure		Arrival	
City	Time	City	Time
B	8:00 am	A	3:00 pm
A	4:00 pm	B	8:00 pm

Assume that planes cruise at the same speed in both directions. However, the effective speed is influenced by a steady wind blowing from east to west at 50 km per hour.

12. What is the time difference between A and B?

(1) 1 hour and 30 minutes

(2) 2 hours

(3) 2 hours and 30 minutes

(4) 1 hour

13. What is the plane's cruising speed in km per hour?

Question 14

The speed of an escalator that is going down from the first floor to the ground floor is 3 steps/s. A is coming down from the first floor at 7 steps/s. B is going up from the ground floor at 10 steps/s. They meet in 5 s, on the same step. How many steps are there in a stationary escalator?

(1) 84

(2) 85

(3) 86

(4) 87

Question 15

Sanjay takes 30 steps in 20 s to come down an escalator. Vijay takes 40 steps in 10 s to come down the same escalator. How many steps are there in a stationary escalator?

(1) 45

(2) 50

(3) 60

(4) 35

Question 16

A man can walk up in a moving escalator (upwards) in 30 s. The same man can walk down this moving 'up' escalator in 90 s. Assume that his walking speed is same in both upward and downward motion. How much time will he take to walk up the escalator when it is not moving?

(1) 30 s

(2) 75 s

(3) 60 s

(4) 45 s

Question 17

An escalator is moving at 3 steps per second. Harish walks in the same direction as the moving escalator at 2 steps per second and takes 3 second less to get out of the escalator than when he was moving on the stationary escalator. How many steps are there in the stationary escalator?

Question 18

Rajat takes 11 steps to reach the bottom of an escalator in 35 seconds. Sharad walks on the same escalator and takes 39 steps in 21 sec to reach the bottom. How many steps are visible on the escalator when it is not moving? (Note: The escalator is coming down in both the cases)

Question 19

An escalator is coming down at a constant speed. Vivek takes 40 steps to reach to the bottom. The time in which he takes 15 steps, Samar manages to reach the bottom by taking 45 steps. If the escalator were turned off, how many steps would they have to take to walk down?

Question 20

Ravi and Bhanu are walking on a moving escalator, with Ravi walking in the same direction as the escalator and Bhanu walking in the opposite direction. Ravi's speed is 5 steps per second, and Bhanu's speed is 10 steps per second. If they start at either end of the escalator simultaneously and meet 10 seconds after they started, how many steps are there on the escalator?

(1) 140

(2) 145

(3) 150

(4) 160

Thanks!